

StepperTune

User Manual

MIDI to G-code music engine

1. Introduction

StepperTune converts melodies written manually or imported from MIDI files into G-code for start-of-print and end-of-print jingles on supported Bambu Lab printers. The program lets you preview the melody on your computer and automatically applies the correct driver for the selected printer model.

Important: StepperTune generates the musical section to be inserted into the printer preset. It does not replace the normal print G-code.

2. Interface

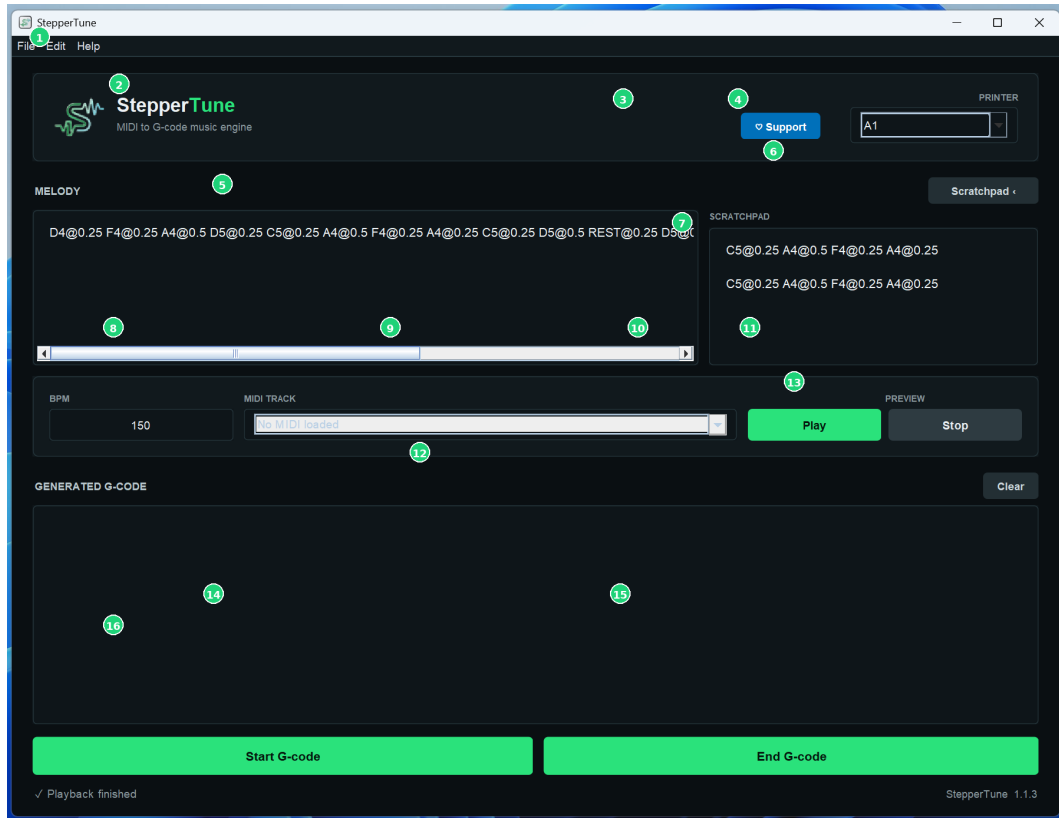


Figure 1 - Main interface with the Scratchpad open and numbered references.

Ref.	Control	Function
1	Main menu	Access to File, Edit and Help.
2	Logo and identification	Program name and description.
3	Support	Opens the project support/donation link.
4	Printer	Selects the printer model.
5	Melody	Main melody editor. This is the source used to generate G-code.
6	Scratchpad	Shows or hides the temporary workspace.
7	Scratchpad area	Stores, edits and previews fragments without including them in generated G-code.
8	BPM	Melody tempo in beats per minute.
9	MIDI Track	Selects the imported MIDI track.
10	Play	Previews the active text area.
11	Stop	Stops the preview.
12	Generated G-code	Displays the generated G-code.
13	Clear	Clears the G-code output.
14	Start G-code	Generates the start-of-print jingle.
15	End G-code	Generates the end-of-print jingle.
16	Status bar	Displays status messages and operation confirmations.

3. Creating a melody

Notes can be typed directly into the Melody area. Each event uses the format **Note@Duration**, and events are separated by a space.

```
C4@0.25 D4@0.25 E4@0.5 REST@0.25 G4@0.5
```

3.1 Notes, octaves and rests

- Available notes: C, C#, D, D#, E, F, F#, G, G#, A, A#, B.
- The number after the note indicates the octave, for example C4 or F#5.
- REST inserts a pause and uses the same duration syntax, for example REST@0.25.
- Note names are not case-sensitive.

3.2 Duration and BPM

Values such as 1, 0.5, 0.25 and 0.125 represent relative musical durations. BPM determines the actual playback speed of the sequence.

3.3 Selection and preview

Melody and Scratchpad share the same Play and Stop controls. Only one text area is active for playback: selecting notes in one area clears any selection in the other.

- Double-click a single note: the note is selected. Press Play to start playback from that note and continue forward.
- Select two or more consecutive notes: press Play to preview only the selected notes.
- With no selection, Play starts from the beginning of the active text area.

During preview, a highlighted playback cursor follows the current note and the view scrolls automatically to keep upcoming notes visible.

3.4 Scratchpad

The Scratchpad is a temporary workspace that stays hidden when it is not needed. It can contain one or more melody fragments and supports normal Cut, Copy and Paste operations while manually arranging a melody.

Important: Scratchpad content can be previewed, but it is **never used to generate G-code**. Only the Melody area is used as the source for G-code generation.

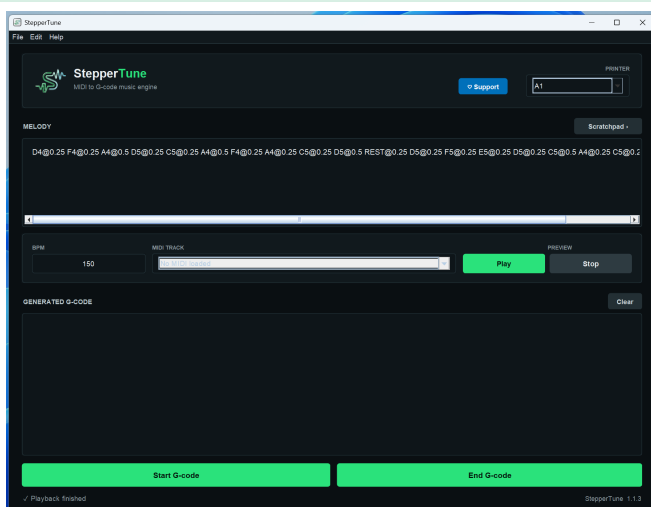


Figure 2 - Scratchpad closed.

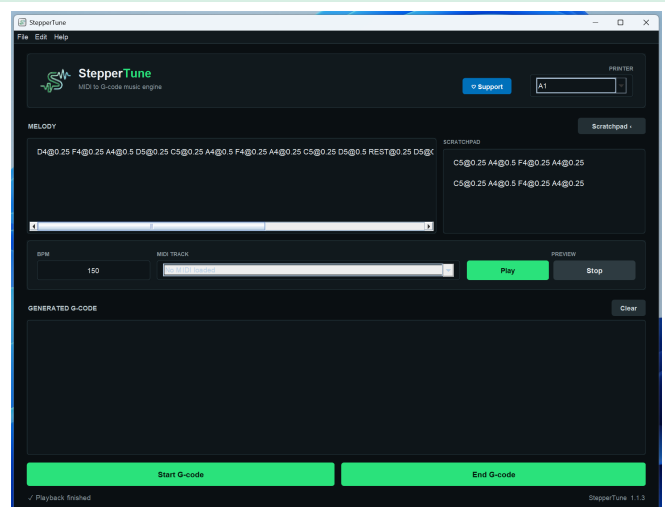


Figure 3 - Scratchpad open.

4. Managing a StepperTune project

StepperTune uses the native **.stf** format to save your work and continue it later. The project file stores the BPM value, the main melody and the Scratchpad contents.

4.1 New File

File > New File creates a new project: it clears Melody, Scratchpad and Generated G-code, resets the BPM value, and clears MIDI and playback-selection state.

4.2 Open Project...

File > Open Project... opens an **.stf** file and restores the saved Melody, Scratchpad and BPM.

4.3 Save Project...

File > Save Project... saves the current project as an **.stf** file. Scratchpad content is preserved even when the Scratchpad panel is hidden.

5. Importing a MIDI file

1. Open File > Load MIDI... and select a **.mid** file.
2. If the MIDI contains multiple tracks, choose the desired one in the MIDI Track field.
3. Check the converted melody and the detected BPM.
4. Press Play to preview the result before generating G-code.

With complex MIDI files, the selected track can greatly affect the result. Tracks with a clear melodic line and little overlap generally work best.

6. Preview, generation and export

1. Select the correct printer model in the Printer field.
2. Prepare the melody manually or import it from MIDI.
3. Use Play and Stop to check the preview. If multiple consecutive notes are selected, Play previews only the selection.
4. Press Start G-code for the start-of-print jingle or End G-code for the end-of-print jingle.
5. The result appears in Generated G-code.
6. Use File > Export G-code... to save the result as a **.gcode** file, or copy it manually.

Note: the preview uses your computer audio. The actual printer sound is produced by stepper motor vibration and will therefore sound different.

7. Adding the jingle in Bambu Studio

The following procedure is adapted from the official Bambu Lab documentation for MIDI/G-code sounds.

[Bambu Lab Wiki - MIDI](#)

7.1 Select the printer and preset

In Bambu Studio, open the Prepare tab and select the printer you are using from the Printer drop-down list, including the correct nozzle size. Click the Edit preset button next to the printer name to open Printer settings.

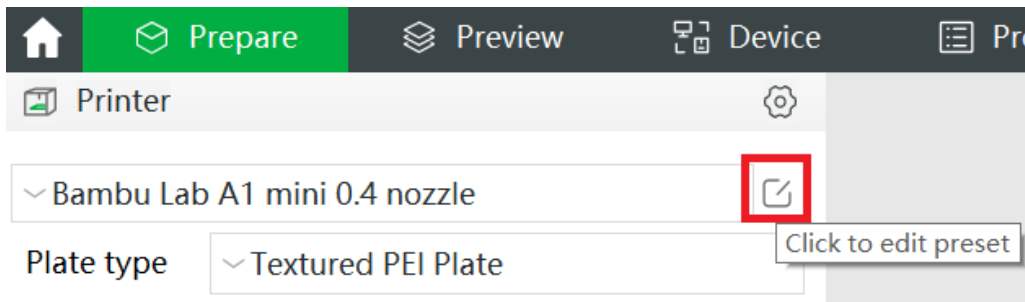


Figure 4 - Opening the printer preset. Image source: Bambu Lab Wiki.

7.2 Enable Advanced and open Machine G-code

In Printer settings, enable the Advanced switch in the upper-right corner, then open the Machine G-code tab.

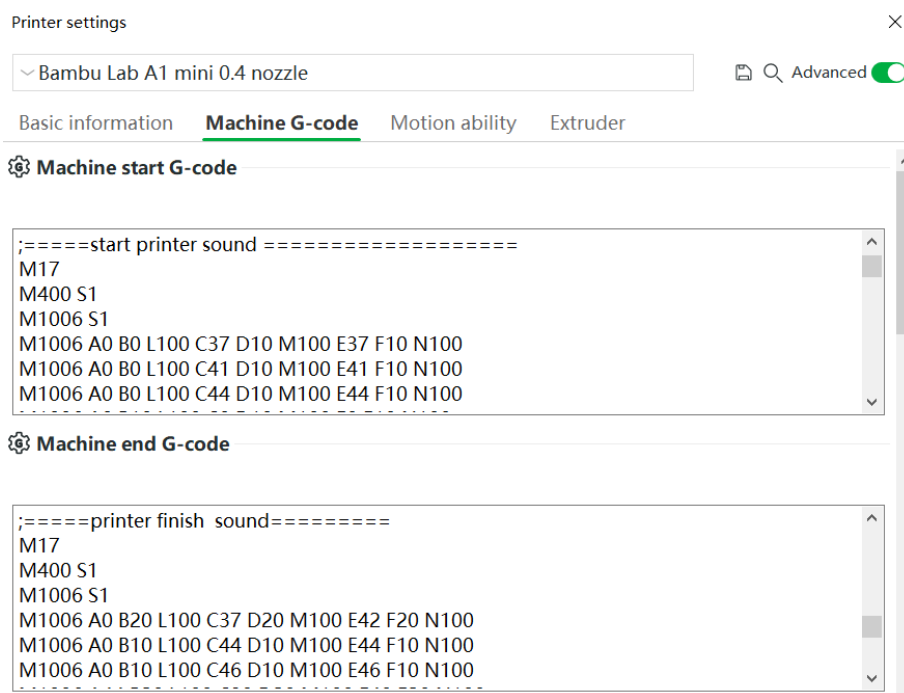


Figure 5 - Machine G-code tab. Image source: Bambu Lab Wiki.

7.3 Edit the G-code

To set the start sound, in **Machine start G-code** find the **start printer sound** divider and replace **all the code between this divider and the next one** with the G-code generated by **Start G-code** in StepperTune.

To set the end sound, in **Machine end G-code** find the **printer finish sound** divider and replace **all the code between this divider and the next one** with the G-code generated by **End G-code** in StepperTune.

NOTE: Preserve the line breaks in the generated code when pasting G-code. The official Bambu Lab guide warns that losing line breaks may cause code errors. On Windows, it recommends opening the G-code file with **WordPad** before copying and pasting it into Bambu Studio; using **Notepad** may cause line break loss.

7.4 Save the preset

Click **Save** in the upper-right corner to save the current preset. You can save it as a **User Preset** or a **Project Inside Preset**, depending on how often you want to use the customized jingle.

If saved as a **User Preset**, the preset can be used by all projects and the printer will play the customized start/end sound whenever that preset is selected. If saved as a **Project Inside Preset**, the customized sound will only be used when that specific project is printed.

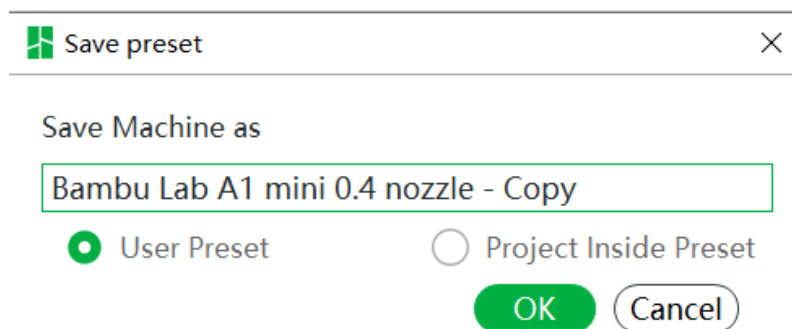


Figure 6 - Saving the preset. Image source: Bambu Lab Wiki.

8. Supported printers

- A1
- A1 mini
- A2L
- H2S
- H2D
- X2D
- P2S

Always select the printer model you are actually using. StepperTune automatically applies the sound parameters and G-code structure defined by the corresponding driver.

9. Usage tips

- Start with short, recognizable jingles.
- Always listen to the preview before generating G-code.
- Use the Scratchpad to temporarily store fragments while composing or rearranging notes.
- With complex MIDI files, try several tracks and choose the cleanest one.
- Save your project periodically in .stf format while composing.
- Keep a copy of the printer's original preset before modifying it.
- After making a change, perform an initial print or test under controlled conditions.

10. Responsibility and safety

StepperTune is an independent project and is not an official Bambu Lab product. It is not affiliated with, sponsored by, or endorsed by Bambu Lab.

The software generates G-code commands that operate the printer motors to produce sounds. The user is responsible for verifying the selected printer model, the generated content and its correct insertion into the printer preset.

Use of manually modified G-code, use on unsupported printer models, or replacement of Machine G-code sections other than the sound block is entirely at the user's own responsibility.

Firmware, Bambu Studio and G-code command behavior may change over time. Always verify correct operation after printer firmware or software updates.

The author of StepperTune cannot be held responsible for printer damage, failed prints, data loss or other effects resulting from improper use, manual modifications or use on unsupported hardware.

11. References

Bambu Studio procedure and reference images: [Bambu Lab Wiki - MIDI](#).

Bambu Lab, Bambu Studio and the printer model names mentioned belong to their respective owners.